

Ethical Work as Profitable Work: The Alignment Theorem Version 2 for Policy-Governed Networks and LLM Agents

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Abstract

We study a direct idea: LLM agents on a network can be trained to be ethically aligned and profit-seeking, automating useful work while maintaining alignment. If the ethical path is the most profitable path, the network can run more smoothly. To make this idea testable, “ethical” is defined relative to an explicit, versioned community policy over observable actions. A trained agent may propose actions and pursue profit, while a deterministic policy gate decides what the network may publish.

Version 2 replaces the unbounded scarcity argument of Version 1 with a finite incentive theorem. If the funded reward advantage plus conservative expected enforcement exceeds the maximum private deviation gain, the extra cost of compliance, and the agent’s optimization error, then every approximately utility-maximizing choice in the stated model is policy compliant. Separate theorems establish that published actions satisfy every gate obligation, rejected settlements have no effect, and accepted rewards conserve a committed reserve. We mechanize these results in Lean 4.33.0 and execute the policy kernel on Tau Language 0.7.0-alpha at source commit `fd137e860b60083b36f9159ec8090cb1a3c3cb5a`. The result is a finite, replayable foundation for future network states. It guarantees policy-relative behavior at the publication boundary under explicit assumptions; it does not claim objective moral truth, honest unauthenticated sensors, or infallible internal alignment of learned agents.

1 Motivation

Future network states may coordinate markets, public goods, research, dispute resolution, infrastructure, and machine labor. LLM agents could perform much of this work. They can search for opportunities, negotiate, explain proposals, translate human requirements, and execute routine plans. A network that merely asks these agents to maximize profit creates an obvious risk: the easiest profit may come from deception, hidden externalities, duplicated claims, manipulation of evidence, or exploitation of underspecified rules.

The motivating claim of this paper is deliberately stated in human language:

LLM agents on the network can themselves be trained to be ethically aligned and profit-seeking. This can automate the work of the network while maintaining alignment. If the ethical path is the most profitable path, the network can run smoothly.

Training an agent to follow behavioral principles is technically possible; constitutional and constrained-learning methods provide concrete examples [3, 1]. Training alone does not establish a durable network guarantee. Learned systems can encounter distribution shift, exploit proxy rewards,

or tamper with reward channels [2, 4]. We therefore divide the system into two complementary layers:

1. training helps agents find profitable, policy-compliant work; and
2. a deterministic Tau policy gate controls which proposed actions may be published and rewarded.

The learned agent improves automation. The gate owns admission authority.

2 What Version 2 Corrects

Version 1 connected ethical scoring, scarcity, rewards, Tau specifications, and a Lean development. It supplied the important intuition that economics can reinforce community values. Its formal model did not yet establish that intuition.

First, Version 1 defined an action as ethical when an account EETF was at least one, then let a rational choice function select the EETF directly. The action, evidence, and scoring process were missing from the theorem. Second, the payoff function made ethical expected value positive and unethical expected value negative for every positive scarcity value. The Lean theorem therefore selected a scarcity threshold of zero. Scarcity scaled a preference ordering that the definition had already imposed. Third, scarcity canceled from the paper’s printed indifference equation, so increasing scarcity could not move that threshold as claimed. Fourth, asymptotic real-valued supply did not refine the finite modular arithmetic executed by Tau bitvectors.

Version 2 makes five changes:

1. it reasons about observable actions and authenticated evidence;
2. it replaces infinite scarcity with a finite strict incentive margin;
3. it separates hard admission constraints from soft reward scores;
4. it uses integer atoms and reserve-backed settlement; and
5. it states the host, Tau, Lean, training, and governance boundaries independently.

Scarcity may still be studied as a bounded economic variable. It is no longer the source of the theorem.

3 Policy-Relative Ethics

3.1 Why the policy must be explicit

Tau can reason about formal opinions, agreement, contradiction, and software requirements. It cannot infer physical truth from untrusted observations or decide which human values are objectively correct [7]. Version 2 consequently uses a policy-relative definition.

Definition 1 (Network policy). *At epoch t , Φ_t is a versioned formal policy selected under the network’s governance rules. Its identity includes a canonical policy root and the rules under which it may be revised.*

Definition 2 (Policy-compliant action). *Given state s , action a , and evidence w , the action is policy compliant at epoch t when the authenticated verifier and policy jointly accept:*

$$C_t(s, a, w) = 1.$$

This definition does not reduce ethics to whatever a majority wants. It states the exact object the network can enforce. The governance design must still address rights, participation, minority protection, expertise, conflicts, capture, and amendment. Tau’s own design allows participants to choose their consensus rules rather than assuming one universally optimal mechanism [7].

3.2 Hard constraints before soft scores

A scalar score can hide a forbidden act behind compensating benefits. Version 2 therefore represents the policy result conceptually as

(policy root, hard constraints, soft score, evidence confidence, provenance root).

An action becomes reward eligible only after every hard constraint passes. A soft factor such as EETF may then prioritize among eligible actions. In a future network state, hard constraints might include authenticated ownership, voluntary authorization, no double claim, bounded resource use, transparent fees, and evidence provenance. Communities may add richer ethical clauses through governed revisions.

4 System Architecture

The Version 2 path is

human or LLM proposal $\rightarrow$ authenticated evidence $\rightarrow$ Tau policy gate $\rightarrow$ reserve settlement $\rightarrow$ publication.

The action may be proposed by an untrusted optimizer. The publication path checks seven obligations:

1. the active policy root matches;
2. the evidence is authenticated;
3. the action type is registered;
4. the action satisfies the active policy;
5. its nonce is fresh;
6. its task or reward nullifier is unclaimed; and
7. the requested reward is funded.

The gate output is the conjunction of these obligations. Unknown inputs fail closed. A rejected action receives no payout and leaves the reserve unchanged.

Tau Net’s current Alpha Testnet uses an extralogical API for functionality not yet expressible in Tau Language [6]. Version 2 treats that API as a visible adapter boundary. Signature verification, state lookup, evidence authentication, and exact monetary calculations must produce authenticated facts. Tau checks the declared logical relationship among those facts. A caller-supplied Boolean is not an authenticated fact merely because Tau reads it.

5 Finite Incentive Model

Consider agent i choosing between a compliant action a_c and any noncompliant action a_n within the modeled action class. Define:

Symbol	Meaning
R_c	funded reward for the compliant action
R_n	reward still obtainable by a noncompliant action
pL	conservative expected enforcement amount
G_i	maximum private gain from deviation
C_i	maximum additional cost of compliance
ϵ_i	bounded optimization error

All monetary terms are finite. In executable accounting they are nonnegative integer atoms. The product pL may be represented by an exact rational in the reference model or by a conservative integer lower bound in settlement logic.

Assumption 1 (Bounded deviation). *For every modeled noncompliant action, its private advantage over a compliant alternative is at most $G_i + C_i$.*

Assumption 2 (Authenticated enforcement). *The probability and charge summarized by pL are conservatively estimated and cannot be selected by the acting agent.*

Assumption 3 (Funded rewards). *R_c is paid only from a committed reserve, and duplicate claims are rejected.*

Theorem 1 (Finite Alignment Theorem). *If*

$$R_c - R_n + pL > G_i + C_i + \epsilon_i,$$

then no modeled noncompliant action is ϵ_i -optimal. Every ϵ_i -optimal choice is policy compliant.

Proof. Let U_c be compliant utility and U_n the utility of any modeled noncompliant action. The deviation and enforcement assumptions give

$$U_c - U_n \geq R_c - R_n + pL - G_i - C_i.$$

The strict-margin premise implies $U_c - U_n > \epsilon_i$. Hence $U_n + \epsilon_i < U_c$, contradicting the definition of an ϵ_i -optimal noncompliant choice. $\square$

Strictness matters. At equality, a noncompliant choice can remain within the optimizer's permitted error. The executable boundary test retains this counterexample.

Proposition 1 (Reward-only alignment). *Set $pL = 0$. If*

$$R_c - R_n > G_i + C_i + \epsilon_i,$$

then every ϵ_i -optimal modeled choice is policy compliant. Thus the network need not punish agents when the funded advantage of ethical work alone clears the complete deviation bound.

Proposition 2 (Network equilibrium consequence). *If the strict-margin assumption holds for each agent and every modeled noncompliant unilateral deviation at a profile, then no agent has an ϵ_i -optimal noncompliant deviation from that profile. When it holds at every reachable profile, policy compliance is a dominant action class in the restricted game.*

This proposition captures the sense in which the network can run smoothly: routine profit seeking and policy compliance point in the same direction. It does not remove congestion, disagreement, implementation bugs, or unmodeled external markets.

5.1 Coalitions and sybils

For a coalition K , the analogous sufficient condition is

$$R_c(K) - R_n(K) + p_K L_K > G_K + C_K + \epsilon_K.$$

An individual-agent bound does not imply this coalition bound. Reward claims must therefore bind a unique task or contribution nullifier. Identity splitting must not multiply a fixed reward. Version 2’s executable gate includes a `taskUnclaimed` obligation; richer coalition bounds remain future work.

6 Safety and Accounting Theorems

The Lean development proves three complementary properties.

Theorem 2 (Admission safety). *If the policy gate admits an action, all seven gate obligations are true. In particular, an admitted action is policy compliant.*

Theorem 3 (Rejected transition no-op). *If settlement rejects, payout is zero and the reward reserve is unchanged.*

Theorem 4 (Reserve conservation). *If settlement accepts, then*

$$reserve_{post} + payout = reserve_{pre}$$

and payout is at most the pre-settlement reserve.

These statements prevent the incentive theorem from assuming an unbacked or unbounded reward. They do not prove the source of reserve funds is itself ethical; that source is another policy obligation.

7 LLM Agents as Ethical and Profit-Seeking Workers

An LLM agent can be trained on the network’s current policy, examples of valid work, rejected counterexamples, and the economic outcomes of accepted actions. Its objective can combine private profit and network payout while treating hard policy clauses as constraints:

$$\max_{\theta} \mathbb{E}_{a \sim \pi_{\theta}} [\text{private profit}(a) + \text{funded payout}(a)] \quad \text{subject to} \quad C_t(s, a, w) = 1.$$

This supports the human claim in three ways.

1. **Training compatibility.** Behavioral principles and profit seeking can coexist in one agent objective. The policy gives the agent a precise description of acceptable work.
2. **Economic reinforcement.** A profitable compliant path creates repeated evidence that following the policy is useful rather than merely a restriction.
3. **Automated coordination.** Agents can search, negotiate, and complete tasks continuously, while each resulting action remains subject to the same publication gate as a human action.

Present-day LLM assistants are constructive examples of the basic point: an agent can pursue an assigned goal while following ethical and safety constraints. An assistant that completes useful work within those constraints shows that the two behaviors can coexist. This establishes feasibility at the behavioral level. It does not prove that every model will generalize safely to every situation. Version 2 does not require a particular laboratory, model architecture, or training algorithm. Any lab may train an agent against the public policy and receipt interface.

The runtime shield is retained because a capable model may discover behavior outside its training distribution or exploit a proxy reward. The model never grades its own evidence, chooses the active policy root, marks its own task unclaimed, or writes the reward reserve. Training improves how often useful proposals pass. The gate determines what can be committed.

8 Tau Language Implementation

The policy file `tau/v2/alignment_policy_gate_v2.tau` uses current Tau stream declarations and the simple Boolean-function type. It computes

```
allow = policy_root_ok ∧ evidence_authenticated ∧ action_known ∧
       action_policy_compliant ∧ nonce_fresh ∧ task_unclaimed ∧ reward_funded.
```

The replay packet contains the complete 128-row truth table. Its first eight rows are one all-true acceptance followed by one row that falsifies each obligation independently. On Tau Language 0.7.0-alpha commit `fd137e8`, exactly one row was accepted and all 127 other rows were rejected, matching the expected output. The receipt binds the Tau binary and complete packet SHA-256 hashes. Tau remains alpha software [5]; this result is compatibility evidence for the bounded kernel, not a production or mainnet claim.

The simple Boolean kernel is intentional. Tau bitvector arithmetic is modular, including addition, subtraction, multiplication, and division [5]. Version 2 keeps exact integer accounting in the reference settlement model and passes a verified `reward_funded` fact to the Tau policy. A future bitvector settlement kernel must prove every no-wrap precondition and preserve parity with the integer model.

9 Lean Mechanization

The proof project under `proofs/v2` is pinned to Lean 4.33.0 and depends only on the Lean standard library. The main declarations are:

- `admits_implies_every_obligation`;
- `committed_action_is_policy_compliant`;
- `rejected_settlement_is_noop`;
- `accepted_settlement_conserves_reserve`;
- `accepted_payout_is_funded`;
- `epsilon_optimal_choice_is_compliant`; and
- `finite_alignment_theorem`; and

- `reward_only_alignment_theorem`.

Unlike Version 1, the repository includes the project configuration and exact toolchain file. The acceptance command is simply `lake build`. A clean placeholder scan is required before describing the theorem as machine checked. The axiom audit permits only the explicitly recorded Lean standard dependencies `propext` and `Quot.sound`; no user-added axiom is admitted.

10 Executable Counterexample Pressure

The exact Python reference model uses nonnegative integer atoms and rational detection probabilities. Its tests retain the following negative cases:

- equality at the incentive boundary does not imply alignment;
- zero detection can make even a large nominal slash irrelevant;
- every individually false gate obligation rejects without an effect;
- zero, one-atom, exact-reserve, and maximum-u64 neighbors are checked;
- booleans masquerading as monetary integers and out-of-range amounts are rejected;
- a high-profit forbidden proposal cannot pass the policy shield;
- no admissible proposal produces a safe no-op; and
- tie-breaking among equal-profit admitted actions is deterministic.

An exhaustive small-domain oracle independently compares the implemented strict-margin predicate with direct utility inequalities. Exhaustive coverage of this finite toy domain is evidence for that domain only.

11 Governance and Revision

Policy evolution is essential for a future network state. It also creates an authority risk. A current policy must not silently define its own replacement as ethical merely because the replacement is satisfiable. Version 2 separates an enduring constitution H from revisable policy Φ_t and requires

$$\forall a, \quad \Phi_{t+1}(a) = 1 \implies H(a) = 1.$$

The Lean project names this relation `PreservesConstitution`. A complete deployment needs an authenticated update proposal, participant and voting rules, a policy-root transition, activation time, historical receipt verification, and an emergency process whose own authority is bounded. Preference aggregation must publish its identity assumptions, weights, treatment of missing information, and adversarial manipulation bounds. No generic scalar EETF can supply these properties by itself.

12 Threat Model and Nonclaims

The principal failure modes are:

- **Policy misspecification:** the formal rule omits an important harm or rewards a proxy;
- **evidence forgery:** an adapter marks false evidence as authenticated;
- **unbounded external gain:** the true benefit of deviation exceeds G_i ;
- **governance capture:** an attacker changes the active policy or its identity rules;
- **sybil multiplication:** one contribution receives multiple rewards;
- **coalition deviation:** individually safe margins fail for a cooperating group;
- **model reward hacking:** an agent attacks the grader, evidence path, or reward channel; and
- **implementation drift:** host, Tau, Lean, documentation, and deployed settlement implement different relations.

Version 2 proves neither objective ethics nor the moral correctness of community consensus. It does not infer truth from unauthenticated sensors. It does not prove that an LLM’s internal goals are aligned, that every future action lies in the modeled action class, or that a private deviation bound remains valid forever. It supplies conditional guarantees whose assumptions can be inspected, tested, governed, and revised.

13 Research Program for a Future Network State

The next steps are modular:

1. define a constitutional policy vocabulary for ownership, consent, evidence, due process, public goods, and bounded externalities;
2. build authenticated adapters for signatures, proofs, oracles, and task completion receipts;
3. prove host–Tau–Lean parity for each policy release;
4. measure realistic private deviation gains rather than assuming them;
5. develop coalition, sybil, bribery, and governance-capture bounds;
6. train agents from any capable laboratory against versioned policy and receipt interfaces; and
7. run them first in shadow mode, preserving every divergence and reward hacking attempt as negative evidence.

This program permits increasingly capable agents without granting them publication authority. As agents improve, they should become better at finding profitable ethical work. As policy improves, the same interface gives those agents a clearer target.

14 Conclusion

Ethics and profit need not be designed as permanent adversaries. A network can state the behavior it will accept, fund valuable compliant work, bound the advantage of deviation, and train automated agents to search within that space. When the finite incentive margin is strict, policy compliance is the profitable choice in the stated model. When the publication gate is enforced, arbitrary proposals cannot bypass the policy merely because an agent expects a larger private gain.

The practical vision is unchanged from the motivating idea: ethically aligned, profit-seeking LLM agents can automate much of a future network state’s work and help it run smoothly. Version 2 gives that vision a narrower and stronger foundation: explicit community policy, authenticated evidence, finite economics, reserve-backed rewards, current Tau execution, and reproducible Lean proofs.

References

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